

Support-Graph Reachability and Matrix-Composition Obstructions

Image–Kernel Mismatch with a Rubik-Cube Case Study

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Abstract

Problem. A sector transport graph records whether a projected generator block $Q_i\rho(g)Q_j$ is nonzero. One might infer that a graph path $j \rightarrow k \rightarrow i$ guarantees a nonzero projected composition $Q_i\rho(g_2)Q_k\rho(g_1)Q_j$. That inference is false in general: nonzero adjacent factors can have a zero product.

Approach. We distinguish three objects. The **support graph** records nonzero direct blocks. The **two-step composition graph** records nonzero projected products. The **obstruction set** consists of support-graph paths that disappear under matrix composition. For two adjacent factors $A = Q_i\rho(g_2)Q_k$ and $B = Q_k\rho(g_1)Q_j$, the exact obstruction is $\text{im } B \subseteq \ker A$. We then specialize to representations and sector projectors respecting a common physical-block decomposition.

Results. We prove that the two-step composition graph is a subgraph of the square of the support graph, with equality requiring an additional image–kernel nondegeneracy condition. If the group action and all sector projectors preserve the same physical blocks, projected composition cannot cross between sectors with disjoint block support at any finite length. In the canonical 228-dimensional Rubik realization, five distinguished pairs selected from two-step support paths all satisfy this block obstruction. Their adjacent factor norms are order one, while exhaustive products over the 18^2 ordered generator pairs have Frobenius norms between 1.10×10^{-16} and 3.02×10^{-15} .

Boundary. Graph reachability and projected-composition reachability are distinct invariants. A hybrid sector may be incident to edges carried by different physical-block components without transferring amplitude between those components. The canonical five pairs are therefore not compositional morphisms; they are finite witnesses of support-graph overapproximation. Image–kernel compatibility is the missing promotion datum.

1 Introduction

Sectorized operator analysis frequently begins with a family of orthogonal projectors

$$I = \sum_{i \in \mathcal{I}} Q_i, \quad Q_i Q_j = \delta_{ij} Q_i,$$

and a unitary action $\rho : G \rightarrow U(V)$. For every declared generator g , define the transport blocks

$$T_{ij}(g) = Q_i \rho(g) Q_j.$$

The direct support relation $j \rightarrow i$ is present when one of these blocks is nonzero. This relation is useful: it records exactly where a single generator fails to preserve the chosen sectorization. It does not, however, retain the images and kernels of the nonzero blocks.

This loss of information matters under composition. Given nonzero blocks

$$Q_k \rho(g_1) Q_j \neq 0, \quad Q_i \rho(g_2) Q_k \neq 0,$$

the graph contains a path $j \rightarrow k \rightarrow i$. The corresponding operator is

$$Q_i \rho(g_2) Q_k \rho(g_1) Q_j.$$

The product can vanish if the first step lands entirely within directions annihilated by the second. Thus, for a fixed sectorization and declared generator set, Boolean support-graph reachability overapproximates projected-composition reachability defined by compatible ordered generator tuples.

This paper makes that distinction formal through named definitions, exact obstruction theorems, local promotion criteria, and a promotion problem.

1.1 Scope

The general theorems concern finite-dimensional Hilbert spaces, orthogonal sector projectors, and unitary or linear transport maps. The Rubik’s cube representation supplies one finite realization. Claims about its nine sectors, ten direct edges, and five graph-only pairs are computational certificates for the declared representation and generator set.

This paper does not claim that every support path is obstructed. It identifies the missing datum and provides exact local criteria for both obstruction and promotion. Higher-depth accessibility theories require additional declared objects and are not considered here. The criterion supplied here can be used as the local survival gate in any later routed-composition analysis.

Notation

Symbol	Meaning
V	finite-dimensional complex Hilbert space
$\rho : G \rightarrow U(V)$	declared transport representation
$S \subset G$	finite generator set; inverse-closed in the canonical audit
Q_i	orthogonal projector onto sector $V_i = \text{im } Q_i$
$T_{ij}(g)$	direct transport block $Q_i \rho(g) Q_j$
K_{ij}^S	$\max_{g \in S} \ T_{ij}(g)\ _F$
$\mathcal{G}_S$	directed support graph of the direct blocks
$C_{ikj}(g_2, g_1)$	projected two-step composition $Q_i \rho(g_2) Q_k \rho(g_1) Q_j$
$\mathcal{G}_{\text{comp}}^{(2)}$	two-step composition graph
$\mathcal{O}_{\text{wit}}^{(2)}$	witness-level image–kernel obstruction tuples
$\mathcal{O}_{\text{path}}^{(2)}$	path-level triples whose every declared two-step product vanishes
Π_b	orthogonal projector onto physical block V_b
$\text{BlockSupp}(Q_i)$	blocks on which Q_i has nonzero restriction

2 Sector Transport and the Support Graph

2.1 Sectorization

Let V be finite dimensional and let $\{Q_i\}_{i \in \mathcal{I}}$ be a complete orthogonal sectorization:

$$Q_i = Q_i^\dagger = Q_i^2, \quad Q_i Q_j = 0 \ (i \neq j), \quad \sum_i Q_i = I.$$

We do not assume that Q_i commutes with $\rho(G)$. Indeed, non-invariance is the source of off-diagonal direct transport.

2.2 Support Graph

Definition 2.1 (Direct support graph).

For a declared generator set S , the directed support graph $\mathcal{G}_S$ has vertex set $\mathcal{I}$ and edge

$$j \longrightarrow i \iff T_{ij}(g) = Q_i \rho(g) Q_j \neq 0 \text{ for some } g \in S.$$

The associated weighted support matrix is

$$K_{ij}^S = \max_{g \in S} \|Q_i \rho(g) Q_j\|_F.$$

When $S = S^{-1}$ and ρ is unitary, K^S is symmetric, although the block-level definition remains directed.

2.3 Transport–Non-Invariance Identity

The support graph has the following exact operator interpretation.

Theorem 2.2 (Transport–Non-Invariance Identity).

For a unitary U and an orthogonal projector Q_j in a complete sectorization,

$$\sum_{i \neq j} \|Q_i U Q_j\|_F^2 = \|(I - Q_j) U Q_j\|_F^2 = \frac{1}{2} \| [U, Q_j] \|_F^2.$$

Proof. Orthogonality gives $I - Q_j = \sum_{i \neq j} Q_i$, so the first equality is the Pythagorean decomposition of the outgoing block. Put $Q = Q_j$. Relative to $V = \text{im } Q \oplus \text{ker } Q$, the commutator has off-diagonal blocks $(I - Q)UQ$ and $-QU(I - Q)$. Unitarity gives $\|(I - Q)UQ\|_F^2 = \text{tr}(Q) - \|QUQ\|_F^2$ and $\|QU(I - Q)\|_F^2 = \text{tr}(Q) - \|QU^\dagger Q\|_F^2$. The two compressed operators are adjoints, so $\|QUQ\|_F = \|QU^\dagger Q\|_F$. Hence the two off-diagonal blocks have equal Frobenius mass, and their squared norms sum to $\|[U, Q]\|_F^2$. $\square$

Corollary 2.3.

Vertex j has no outgoing off-diagonal edge for U if and only if $[U, Q_j] = 0$.

This result is local to one generator. It does not assert that paths in the resulting support graph compose nontrivially.

3 Projected Matrix Composition

3.1 Composition Operator

Definition 3.1 (Two-step composition operator).

For sectors j, k, i and generators $g_1, g_2 \in S$, define

$$C_{ikj}(g_2, g_1) = Q_i \rho(g_2) Q_k \rho(g_1) Q_j.$$

The ordered triple (j, k, i) is **composition-active** if this operator is nonzero for at least one ordered pair (g_1, g_2) .

Definition 3.2 (Two-step composition graph).

The graph $\mathcal{G}_{\text{comp}}^{(2)}$ contains $j \Rightarrow i$ when there are $k \in \mathcal{I}$ and $g_1, g_2 \in S$ such that $C_{ikj}(g_2, g_1) \neq 0$.

The graph square $\mathcal{G}_S^2$ contains $j \rightsquigarrow i$ when there is an intermediate k with edges $j \rightarrow k$ and $k \rightarrow i$. The two relations are deliberately different.

3.2 Composition-to-Path Inclusion

Theorem 3.3 (Composition-to-Path Inclusion).

If $C_{ikj}(g_2, g_1) \neq 0$, then

$$Q_k \rho(g_1) Q_j \neq 0 \quad \text{and} \quad Q_i \rho(g_2) Q_k \neq 0.$$

Consequently,

$$\mathcal{G}_{\text{comp}}^{(2)} \subseteq \mathcal{G}_S^2.$$

Proof. A product with a zero factor is zero.

The converse fails. This is not a numerical artifact but a direct consequence of ordinary matrix algebra. $\square$

Definition 3.4 (Witness-level obstruction set).

The witnessed obstruction set is

$$\mathcal{O}_{\text{wit}}^{(2)} = \{(j, k, i, g_2, g_1) : Q_k \rho(g_1) Q_j \neq 0, Q_i \rho(g_2) Q_k \neq 0, C_{ikj}(g_2, g_1) = 0\}.$$

Definition 3.5 (Path-level obstruction set).

The graph-only path obstruction set is

$$\mathcal{O}_{\text{path}}^{(2)} = \{(j, k, i) : j \rightarrow k \rightarrow i \text{ in } \mathcal{G}_S, C_{ikj}(g_2, g_1) = 0 \forall g_1, g_2 \in S\}.$$

The Image–Kernel Criterion acts on a fixed witness tuple in $\mathcal{O}_{\text{wit}}^{(2)}$. The canonical five-pair census is a path-level statement in $\mathcal{O}_{\text{path}}^{(2)}$.

4 Exact Obstruction Theorems

4.1 Image–Kernel Criterion

Theorem 4.1 (Image–Kernel Criterion).

Fix i, k, j and g_1, g_2 . Set

$$A = Q_i \rho(g_2) Q_k, \quad B = Q_k \rho(g_1) Q_j.$$

Then

$$C_{ikj}(g_2, g_1) = AB = 0 \iff \text{im } B \subseteq \ker A.$$

In particular, $A \neq 0$ and $B \neq 0$ do not imply $AB \neq 0$.

Proof. The product AB vanishes exactly when $A(Bv) = 0$ for every v , which is equivalent to every vector in $\text{im } B$ belonging to $\ker A$.

This theorem identifies the information discarded by the support graph. A tuple belongs to $\mathcal{O}_{\text{wit}}^{(2)}$ exactly when both factors are nonzero and the displayed image–kernel inclusion holds. An edge records only that a block is nonzero; it does not record its image subspace or the kernel of the next block. $\square$

4.2 Composition Mass and Efficiency

For a fixed path and generator pair, define

$$M_{ikj}(g_2, g_1) = \|C_{ikj}(g_2, g_1)\|_F$$

and, when both factors are nonzero,

$$\eta_{ikj}(g_2, g_1) = \frac{\|AB\|_F}{\|A\|_F \|B\|_F}.$$

Submultiplicativity gives $0 \leq \eta \leq 1$. While the support graph records only that the denominator is nonzero, the quantity η measures the extent to which the two incident blocks are compositionally aligned.

4.3 Disjoint Endpoint Block-Support Obstruction

Suppose now that

$$V = \bigoplus_{b \in \mathcal{B}} V_b, \quad I = \sum_b \Pi_b,$$

and that both the action and sectorization respect this decomposition:

$$[\rho(g), \Pi_b] = 0, \quad [Q_i, \Pi_b] = 0$$

for every g, i, b . Write

$$Q_i = \bigoplus_b Q_i^{(b)}, \quad \text{BlockSupp}(Q_i) = \{b : Q_i^{(b)} \neq 0\}.$$

Theorem 4.2 (Disjoint Endpoint Block-Support Obstruction).

If

$$\text{BlockSupp}(Q_i) \cap \text{BlockSupp}(Q_j) = \emptyset,$$

then for every finite sequence of intermediate sectors and group elements,

$$Q_i \rho(g_n) Q_{k_{n-1}} \cdots Q_{k_1} \rho(g_1) Q_j = 0.$$

Proof. Every factor is block diagonal, so the full product is block diagonal. Its restriction to block b contains $Q_i^{(b)}$ on the left and $Q_j^{(b)}$ on the right. For every b , at least one of these endpoint restrictions is zero. Hence every block of the product vanishes. $\square$

Corollary 4.3 (Hybrid sectors do not switch blocks).

With the convention $T_{ij} = Q_i \rho(g) Q_j$, vertex j is the source and vertex i is the target. A hybrid intermediate k can have an incoming edge $j \rightarrow k$ supported on one physical-block component and an outgoing edge $k \rightarrow i$ supported on another. Because $Q_k = \bigoplus_b Q_k^{(b)}$ contains no off-diagonal map between these components, such incidence does not promote the path to a cross-block composition.

Corollary 4.4 (All-length obstruction).

Under the hypotheses of Theorem 4.2, graph paths of arbitrary length between disjoint-block endpoints remain operator-inactive after inserting the corresponding sector projectors.

4.4 Local Promotion Criteria

The obstruction theorem gives an exact criterion and two convenient sufficient certificates.

Proposition 4.5 (Exact local promotion criterion).

For $A = Q_i \rho(g_2) Q_k$ and $B = Q_k \rho(g_1) Q_j$,

$$AB \neq 0 \iff \exists v \in \text{im } B \text{ such that } Av \neq 0.$$

Two useful sufficient certificates are:

1. $B \neq 0$ and A is injective on $\text{im } B$;
2. $\text{im } B = \text{im } Q_k$ and $A \neq 0$.

Proof. The equivalence is the negation of Theorem 4.1. For (1), any nonzero vector in $\text{im } B$ survives A . For (2), $A = Q_i \rho(g_2) Q_k$ is nonzero on some vector of $\text{im } Q_k = \text{im } B$.

□

5 The Canonical Rubik Realization

5.1 Declared System

The canonical realization uses

$$V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}},$$

with dimensions $64 + 144 + 8 + 12 = 228$. Every standard face-turn matrix preserves these four cubie-type blocks. The nine QH projectors $Q_1, \dots, Q_9$ are obtained from the registered simultaneous spectral resolution of the numerically commuting operators QT_{all} and HT_{all} . Since those operators are block diagonal, their registered projectors are block diagonal to the declared numerical tolerance.

The registration audit uses the complex128 root-of-unity realization and reports the following maximum Frobenius residuals:

Registration check	Maximum residual
Pairwise commutators of $A_{18}, QT_{\text{all}}, HT_{\text{all}}$	3.77×10^{-16}
Projector idempotence	4.19×10^{-15}
Projector Hermiticity	8.03×10^{-17}
Pairwise projector orthogonality	3.21×10^{-15}
Completeness $\ \sum_i Q_i - I\ _F$	1.32×10^{-14}
Joint invariance under the three QH operators	3.58×10^{-14}
Cross-physical-block projector mass	1.44×10^{-15}

Independent iterative joint diagonalization returns nine sectors with sorted dimensions

$$[1, 2, 8, 20, 26, 27, 39, 39, 66]$$

at clustering tolerances $10^{-6}, 10^{-8}, 10^{-10}$, and 10^{-12} . Although these data certify the registered numerical sectorization, they do not promote it to an exact algebraic joint-spectrum theorem.

The direct transport graph uses the 18 standard face-turn generators. The computational certificate declares three distinct numerical controls:

Control	Value
Direct-edge threshold	5×10^{-2}
Registration and composition-zero tolerance	10^{-10}
Joint-sector clustering sweep	$10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}$

At the declared direct-edge threshold, the graph has ten undirected off-diagonal edges. All ten edges join sectors sharing at least one physical block. The general definitions above remain exact nonzero statements; the threshold is part of this finite numerical realization.

5.2 The Five Graph-Only Pairs

The square of the support graph contains five distinguished two-step paths whose endpoints have disjoint physical-block support:

Endpoints	Endpoint block support	Canonical intermediate
S2–S4	eo vs. ep+co	S6
S3–S9	ep+eo vs. cp+co	S7
S4–S5	ep+co vs. eo	S6
S4–S8	ep+co vs. cp	S9
S6–S9	ep+eo vs. cp+co	S7

These are support-graph statements: they show that the intermediate sector is incident to both endpoint edges, not that it transfers amplitude between the endpoint block components.

5.3 Canonical Composition Audit

Computational Proposition 5.1 (Canonical five obstruction witnesses). In the declared complex128 realization, exhaustive evaluation over all 18^2 ordered generator pairs gives:

Endpoints	Intermediate	Max left edge	Max right edge	Max product (S^2)
S2–S4	S6	0.577350	3.464102	1.098×10^{-16}
S3–S9	S7	3.605551	4.062019	3.017×10^{-15}
S4–S5	S6	3.464102	0.816497	1.553×10^{-16}
S4–S8	S9	1.000000	2.828427	1.060×10^{-15}
S6–S9	S7	3.605551	4.062019	2.937×10^{-15}

The left and right columns are independently maximized over $g_2 \in S$ and $g_1 \in S$. They summarize the individual edge strengths and need not arise from the same ordered generator pair. The final column is the maximum over all $(g_2, g_1) \in S^2$. The edge maxima are order one, while the composition maxima are machine-zero. Theorem 4.2 explains the pattern structurally: the two incident edges use different physical-block components of the intermediate sector.

Claim boundary. The nonzero factor norms and machine-zero products are computational certificates for the declared matrices. Exact vanishing follows conditionally from the exact block-preserving construction and exact block support. No conclusion here depends on a full- G commutant decomposition, a layer-wise isotypic decomposition, or the candidate value $\dim \text{End}_G(V) = 610$.

Support paths versus projected matrix composition

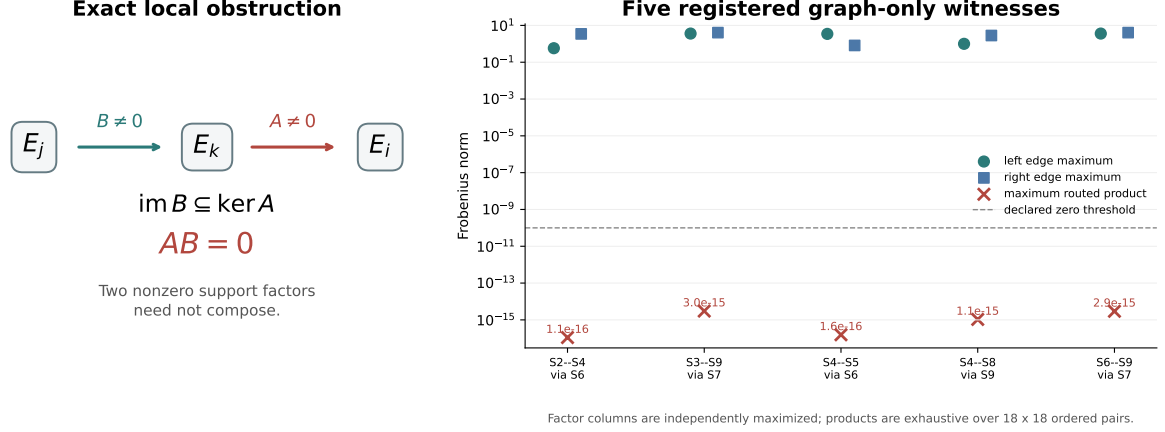


Figure 1. The exact local image–kernel obstruction and the five registered Rubik witnesses. The two factor columns are independently maximized, while the product column is exhaustive over all 18^2 ordered generator pairs. Machine-zero product norms are computational data; exact vanishing uses the stated block-preserving hypothesis.

5.4 Graph-Only Detection Boundary

The graph-level detector applies three tests:

1. the endpoints had disjoint block-support labels;
2. the direct endpoint block was zero;
3. the weighted support graph contained a two-step path.

Although these tests identify the five rows above as graph paths, they do not evaluate projected matrix products. The exhaustive matrix audit supplies the additional conclusion: every declared two-step product for these five path triples is machine-zero and satisfies the structural block obstruction.

6 Support Graphs as Overapproximations

6.1 Boolean Support Forgets Linear Geometry

Replacing each transport block by the Boolean value $\mathbf{1}[T_{ij}(g) \neq 0]$ forgets rank, image, kernel, singular directions, phase, and cancellation. Boolean matrix multiplication therefore computes possible support paths rather than guaranteed nonzero operator products.

This distinction is familiar in sparse linear algebra: the structural pattern of a matrix product can contain entries that cancel or vanish because the corresponding row and column subspaces are orthogonal. Here the loss is stronger because each graph edge aggregates over generators and discards the witness-specific subspace geometry.

6.2 Witness Compatibility

A path $j \rightarrow k \rightarrow i$ may use one generator to establish the first edge and another to establish the second. Even after those witnesses are fixed, composition requires

$$\text{im}(Q_k \rho(g_1) Q_j) \not\subseteq \text{ker}(Q_i \rho(g_2) Q_k).$$

Consequently, a composition-aware data structure must retain more information than K_{ij}^S . Possible choices include:

- the family of block maps $T_{ij}(g)$;
- their image and kernel projectors;
- singular-value data;
- the composition mass tensor $M_{ikj}(g_2, g_1)$;
- normalized composition efficiency $\eta_{ikj}(g_2, g_1)$.

The scalar maximum K^S is sufficient for direct support and insufficient for compositional reachability.

7 Related Work and Novelty Boundary

The support graph is a Boolean abstraction of a family of linear maps. Boolean zero-pattern calculus and combinatorial matrix theory record possible product support, but they do not retain cancellation, image, kernel, or singular-direction data [2, 5]. The inclusion $\mathcal{G}_{\text{comp}}^{(2)} \subseteq \mathcal{G}_S^2$ is the corresponding one-sided abstraction statement for projected sector blocks.

Products selected from a declared family also occur in switched linear systems [6]. The present result is narrower: it does not study asymptotic stability or semigroup growth, but asks when a Boolean path in a sector support graph is realized by a nonzero ordered product. The image–kernel criterion supplies the exact local obstruction, and the block-support theorem supplies an all-length structural obstruction.

Linear representations of free monoids and noncommutative rational series provide a direct comparison for word-indexed matrix products [1]. In this context, the additional sector projectors select a fixed route and can annihilate an otherwise admissible product. The resulting image–kernel condition is also the local matrix incidence underlying the rank-stratified treatment in [3]; that broader incidence geometry is not needed for the criterion proved here.

The underlying image, kernel, singular-value, and block-matrix language is standard matrix analysis [5]. The ambient finite-group language is standard representation theory [7, 4].

8 Graph-to-Composition Promotion Problem

The promotion question is open:

When does a support-graph path force a nonzero projected matrix composition?

In the shared typed notation, the direct relation studied here is $R_1^{\text{op}} = R_1[\rho(S)]$. It is a Boolean overapproximation and cannot be substituted for routed-composition support C_d^{op} , full-word support W_d^{op} , or either associated depth.

For length two, Theorem 4.1 gives the exact answer for fixed witnesses: promotion occurs precisely when the incoming image is not contained in the outgoing kernel. The broader problem is to derive useful hypotheses from coarser data.

Candidate directions include:

1. **Rank conditions.** Determine when ranks of the two incident blocks force nontrivial image overlap inside $Q_k V$.
2. **Generic transversality.** On a declared parameter space, identify when image–kernel incidence is exceptional rather than persistent.
3. **Witness compatibility.** Distinguish paths assembled from unrelated edge-maximizing generators from paths admitting a compatible generator tuple.
4. **Higher length.** Develop recursive image propagation rather than Boolean graph powers.
5. **Compression bounds.** Determine which singular-value or principal-angle summaries are sufficient to certify nonzero composition without storing every block matrix.

Any theorem promoting graph reachability to projected-composition reachability must state one of these nondegeneracy mechanisms explicitly. Graph incidence alone is insufficient.

9 Discussion

The obstruction clarifies the role of hybrid sectors. A hybrid sector remains an important spectral object with nonzero components in several physical blocks, and it may be incident to direct transport edges carried by those components. However, a block-diagonal hybrid projector is not a switch between them; its multiple components coexist as a direct sum.

The canonical five pairs expose this distinction cleanly. Each has two order-one adjacent support edges, while the registered audit finds every projected two-step product to be machine-zero. The block-support theorem explains the vanishing under its exact hypotheses. The support graph therefore describes potential incidence, while operator composition describes compatible amplitude propagation.

This separation gives the logical audit order:

spectral sectors $\longrightarrow$ direct transport blocks $\longrightarrow$ support graph $\longrightarrow$ composition audit.

The arrows indicate construction and audit order, not implication or functional determination. Replacing the direct blocks by their Boolean support discards image–kernel data. Theorem 2.2 characterizes the direct off-diagonal mass, while the passage from graph paths to matrix composition has no unconditional converse.

10 Claim Status and Boundary

The table uses the four claim levels. Refuted implications are listed separately as failure boundaries.

Claim	Status
Transport–Non-Invariance Identity	Theorem
Nonzero projected composition implies a support-graph path	Theorem
Image–Kernel Criterion	Theorem
Disjoint Endpoint Block-Support Obstruction	Theorem
Canonical ten-edge direct support graph	Computational Certificate
Canonical five two-step graph-only pairs	Computational Certificate
Canonical five projected products are machine-zero, with a structural block explanation	Computational Certificate
Generic graph-to-composition promotion	Research Program

The paper’s principal failure boundary is explicit: a support-graph path does not imply nonzero composition. Support data are Boolean incidence data and cannot certify matrix composition without image–kernel or equivalent nondegeneracy information.

11 Conclusion

The support graph and the matrix-composition graph are distinct mathematical objects. For projected blocks

$$A = Q_i \rho(g_2) Q_k, \quad B = Q_k \rho(g_1) Q_j,$$

the exact missing condition is

$$\text{im } B \not\subseteq \ker A.$$

In block-preserving systems, this becomes a structural obstruction: a hybrid intermediate sector cannot transfer amplitude between its distinct physical-block components. The five canonical Rubik paths are finite witnesses of that obstruction. They are visible in the square of the support graph but absent from the projected composition graph.

For the fixed sectorization, generator set, and projected-product semantics used throughout this paper, the central principle is therefore:

Boolean support-graph reachability overapproximates projected-composition reachability.

The next problem is to identify nondegeneracy conditions under which that overapproximation becomes exact.

Appendix A Basis-Level Composition Certificate

Let B_i be an orthonormal basis matrix for $Q_i V$, so $Q_i = B_i B_i^\dagger$. Then

$$Q_i \rho(g_2) Q_k \rho(g_1) Q_j = B_i \left(B_i^\dagger \rho(g_2) B_k \right) \left(B_k^\dagger \rho(g_1) B_j \right) B_j^\dagger.$$

Because left multiplication by B_i and right multiplication by $B_j^\dagger$ preserve nonzero status and Frobenius norm,

$$\|Q_i \rho(g_2) Q_k \rho(g_1) Q_j\|_F = \|(B_i^\dagger \rho(g_2) B_k) (B_k^\dagger \rho(g_1) B_j)\|_F.$$

The implementation evaluates these reduced matrices. This avoids repeated dense 228×228 products without changing the certificate.

Appendix B Longer Paths

For a path $i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_n$, define recursively

$$W_0 = Q_{i_0} V, \quad W_r = Q_{i_r} \rho(g_r) W_{r-1}.$$

The projected composition is nonzero exactly when $W_n \neq \{0\}$. A Boolean support path verifies only that each isolated block map is nonzero on some input; it does not verify that the propagated subspace W_{r-1} avoids the next kernel. This recursion is the natural higher-length replacement for graph powers.

Under the hypotheses of Theorem 4.2, every W_r remains inside the physical blocks present at the source. Therefore no sequence of block-respecting projectors and group elements can connect disjoint-block endpoints.

Appendix C Computational Artifacts

The following repository artifacts support the canonical Rubik certificate. The default directory for C1–C2 is `experiments/paper3/`; C3 is relative to the repository root.

Artifact	Role	Short path
C1	exhaustive graph-versus-composition certificate	<code>validation/composition_obstruction.py</code>
C2	source-addressed completed-run observation	<code>results/composition_obstruction.observation.json</code>
C3	direct graph and composition regression	<code>tests/test_transport.py</code>

C1 reconstructs the nine QH projectors and 18 standard generator matrices, reports commutation, projector, and physical-block residuals, checks the nine-sector census across clustering tolerances 10^{-6} through 10^{-12} , and evaluates all 18^2 ordered products for each canonical triple. From the repository root, run it as `python experiments/paper3/<C1 short path>`; append `-check-result` to verify C2 against its declared source hashes without repeating the matrix calculation.

C2 records parameters, runtime, package versions, Git state, and explicit source hashes. It is a source-addressed run record, not an independent certificate; reproducing the certificate requires a clean full run of C1. C1 supports Proposition 5.3.

All listed artifacts are available in the [RIME repository](#).

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